MagCIL

Multimodal Analysis Group

Institute of Informatics and Telecommunications, NCSR "Demokritos", Athens, Greece

PARTNER PROFILE

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WHO WE ARE

MagCIL is the multimodal deep-learning group of the Institute of Informatics and Telecommunications (IIT) at the National Centre for Scientific Research "Demokritos" (NCSR-D), the largest multi-disciplinary research centre in Greece. Ten people: two tenured researchers, four PhD students and four machine-learning engineers. The group combines fundamental research, applied research in funded projects, and teaching in the MSc in Artificial Intelligence co-organised with the University of Piraeus.

EXPERTISE

- **Speech analytics** — emotion recognition and paralinguistics, speaker diarization, clinical speech (cognitive decline, depression), speech recognition for Greek and other low-resource settings, expressive synthesis and synthetic-speech detection.
- **Music information retrieval and generation** — genre, instrument, playing-technique and regional-style recognition, audio fingerprinting for music-use monitoring, datasets for under-represented repertoires, symbolic and text-to-music generation.
- **Computer vision and image generation** — controllable latent diffusion, self-supervised and few-shot visual representation learning, video understanding, remote sensing.
- **Audio event recognition and acoustic scenes** — sound event detection, urban soundscape monitoring at scale, bioacoustics, few-shot acoustic classes, adversarial robustness of audio models.

Cross-cutting methods: self-supervised and contrastive learning, few-shot learning, diffusion and flow matching, robustness under distribution shift, multimodal fusion.

ROLES WE TAKE IN A CONSORTIUM

- Audio-analysis components delivered with defined interfaces, not prototypes left in a repository.
- Multimodal modelling as the technical core of a work package.
- Robustness and independent evaluation: adversarial testing, benchmarking under noise and distribution shift, synthetic-media detection.
- Dataset design and collection, AI-assisted annotation pipelines, public benchmark releases.
- Dissemination, exploitation and training: open-source releases, public demonstrators, MSc-level teaching.

WORK WITH INDUSTRY

Feasibility studies · custom model development · independent evaluation and benchmarking · prototype integration, containerised for on-premises, private-cloud or edge deployment. Contracted bilaterally through NCSR "Demokritos".

SELECTED PROJECTS

PHAROS AI Factory (EuroHPC JU, 2025–2028) — Greece's national AI Factory. MagCIL contributes to the Language & Culture vertical: a platform for the analysis of Greek traditional music with semantic indexing and natural-language search, sound event recognition services, and song identification by audio fingerprinting.

DeployAI (EU, 2024–2027) — the European AI-on-Demand Platform. MagCIL provides music fingerprinting, speech recognition and audio event recognition services.

MI-TRAP (EU, 2022–2027) — mitigating transport-related air pollution. MagCIL provides real-time soundscape quality assessment and vehicle counting from a single low-cost camera at the project's environmental monitoring stations.

FaRADAI (EU, 2022–2026) — frugal and robust AI for defence. MagCIL worked on semi-automatic data annotation, multimodal fusion for military applications, adversarial robustness in audio and few-shot learning.

DeepFake Awareness (Greek Ministry of Digital Governance, 2026–2027) — a national public-awareness platform on synthetic media.

Earlier: Museek (music monitoring of public performance, with GEA), SmartSubs (real-time subtitling to smartglasses). Full list at magcil.github.io/projects.html.

OPEN SOURCE AND INFRASTRUCTURE

deepaudio-lab / **deepaudio-x** — a no-code, containerised platform for training, evaluating and deploying deep audio classifiers, on-premises or in a private cloud. **pyAudioAnalysis** and **deep_audio_features** — widely used audio analysis libraries. Also amvoc (bioacoustics), audio-few-shot-learning, audio-adversarial-attacks, deep-audio-fingerprinting-benchmark and the GreThE dataset.

Compute through NCSR-D infrastructure and the PHAROS AI Factory (EuroHPC).

WHERE WE PUBLISH

ICASSP, EUSIPCO, IEEE MLSP, ICML, ECCV, NeurIPS workshops, SPIE, and journals in signal processing and multimedia. Selected publications at magcil.github.io/pub.html.

CONTACT

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Legal entity for contracts and grant agreements: NCSR "Demokritos". Participant identification code (PIC) and legal-entity documents on request.